

# RNase-based self-incompatibility in cacti

Karolis Ramanauskas  and Boris Igić 

Department of Biological Sciences, University of Illinois at Chicago, Chicago, IL 60607, USA

Author for correspondence:

Boris Igić

Email: [boris@uic.edu](mailto:boris@uic.edu)

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## Summary

- Approximately one-half of all flowering plants express genetically based physiological mechanisms that prevent self-fertilisation. One such mechanism, termed RNase-based self-incompatibility, employs ribonucleases as the pistil component. Although it is widespread, it has only been characterised in a handful of distantly related families, partly due to the difficulties presented by life history traits of many plants, which complicate genetic research. Many species in the cactus family are known to express self-incompatibility but the underlying mechanisms remain unknown.

- We demonstrate the utility of a candidate-based RNA-seq approach, combined with some unusual features of self-incompatibility-causing genes, which we use to uncover the genetic basis of the underlying mechanisms. Specifically, we assembled transcriptomes from *Schumbergera truncata* (crab cactus or false Christmas cactus), and interrogated them for tissue-specific expression of candidate genes, structural characteristics, correlation with expressed phenotype(s), and phylogenetic placement.

- The results were consistent with operation of the RNase-based self-incompatibility mechanism in Cactaceae.

- The finding yields additional evidence that the ancestor of nearly all eudicots possessed RNase-based self-incompatibility, as well as a clear path to better conservation practices for one of the most charismatic plant families.

## Introduction

Flowering plants display an astonishing array of morphological, physiological and life history traits. Perhaps the most striking feature of this group is their propensity for highly dynamic evolution of breeding systems, suites of traits that affect the frequency of self- and cross-fertilisation. Changes in breeding systems have profound consequences for all other traits, because they modify effective population sizes, the opportunity for recombination, and the distribution of genetic variation in space and time (Holsinger, 2000; Barrett, 2003). These changes alter the prospects for evolution across the entire genome (Uyenoyama, 2000; Wright *et al.*, 2013). Unlike animal species, the vast majority of flowering plant individuals is co-sexual, with hermaphrodite flowers, providing ample opportunity for self-pollination and self-fertilisation. And yet, relatively few predominantly self-fertilise.

Nearly one-half of angiosperms instead strictly avoid self-fertilisation (Raduski *et al.*, 2012). They most commonly obligately cross-fertilise by deploying a genetic system, broadly termed 'self-incompatibility' (SI), which causes otherwise fertile hermaphrodites to recognise and reject their own pollen. A variety of molecular genetic mechanisms can effect SI responses. Among the four best understood mechanisms, three are expressed in relatively small groups, each with a distinct genetic basis, indicating that the basic phenomenon of SI is nonhomologous.

Different mechanisms enforce SI in Brassicaceae, Papaveraceae and Primulaceae, each using unrelated gene families in both pistil and pollen (Nasrallah *et al.*, 1985; Foote *et al.*, 1994; Li *et al.*, 2016). The convergent SI genetic causes imply the possibility of dozens of independently evolved occurrences of SI across angiosperms (Igić & Kohn, 2001; Igić *et al.*, 2008), at a minimum. Nevertheless, one mechanism is both widespread and possibly homologous across eudicots (Igić & Kohn, 2001; Steinbachs & Holsinger, 2002). So far, it is known to occur in five families (Solanaceae, Plantaginaceae, Rubiaceae, Rosaceae, and Rutaceae), each of which employ a member of the T2/S-RNase gene family (S-RNases) as the pistil components, and several S-linked F-box-containing proteins (SLFs) as the pollen components of the self-pollen recognition and rejection response (Anderson *et al.*, 1986; Sassa *et al.*, 1996; Xue *et al.*, 1996; Asquini *et al.*, 2011; Liang *et al.*, 2020).

The significance of the RNase-based SI (RSI) mechanism is underscored by the fact that it is present in both rosoid and asterid clades within eudicots. The most recent common ancestor of the containing group lived well over 100 Ma and its descendants account for approximately two-thirds of all flowering plant species (Soltis *et al.*, 2005). Lineages with RSI are also associated with increased diversification rates (Goldberg *et al.*, 2010). Despite the mounting evidence that the mechanism is homologous across eudicots, some doubts remain about orthology of individual components (Aguilar *et al.*, 2015a; Matsumoto & Tao,

2016). Iterations in different families show significant variation in molecular details, most critically the mode of pollen tube rejection, whether it proceeds by self-recognition (*Prunus*) or nonself-recognition (as in all other cases), as well as the role and number of pollen-expressed genes (Kubo *et al.*, 2010; Fujii *et al.*, 2016). Understanding the distribution of SI and orthology of self-recognition genes (genetic mechanisms) across flowering plants can furnish a critical weight of evidence about the nature of trait homology of this system (Igić & Kohn, 2001; Steinbachs & Holsinger, 2002; Igić *et al.*, 2008) and its influence on diversification rates (Goldberg *et al.*, 2010). Synthesis of such knowledge can be used to precisely detect convergence – independent re-evolution of alternative SI mechanisms – help to identify when, where and how often it occurred, as well as to infer the general factors that cause evolutionary innovation. Finally, a better understanding of SI mechanisms allows manipulations to reduce mate limitation and increase fruit set, with direct applications ranging from plant conservation (Mandujano *et al.*, 2010) to agriculture (Muñoz-Sanz *et al.*, 2020).

Uncovering the molecular genetic basis of SI is generally complicated, expensive and time consuming, which may explain the relative paucity of data across angiosperms, and its discovery in only a handful of plant families since 1986 (Anderson *et al.*, 1986). In particular, wild tropical species and many neglected or underutilised crops are not amenable to classic forward genetic approaches because of long generation times, inconvenient stature or difficult plant care. Some unusual features of RSI can, however, be harnessed to overcome the usual obstacles present in uncovering the genetic bases of traits. First, the wide phylogenetic distribution of RSI indicates that T2/S-RNases are clear pistil-part expression candidates in eudicots with unknown basis of SI (Ramanauskas & Igić, 2017). A subset of members in this gene family can be identified by a set of unique structural features, as well as their phylogenetic relationships (Igić & Kohn, 2001; Steinbachs & Holsinger, 2002; Vieira *et al.*, 2008; Ramanauskas & Igić, 2017). Second, their expression is tissue specific, narrowly confined to styles within pistils, where S-RNases are highly expressed (Xue *et al.*, 1996). Similarly, the pollen-expressed gene candidates are highly tissue specific (Sijačić *et al.*, 2004; Kubo *et al.*, 2010). Third, each individual with a functional RSI system is obligately heterozygous at the S-locus, and negative frequency-dependent selection maintains many S-alleles, because plants with rare S-alleles are compatible with more potential mates (Wright, 1939). Therefore, any hunt for such genes ought to recover heterozygous individuals with highly diverged alleles, often preserving deep ancestral polymorphism (Ioerger *et al.*, 1990). Finally, simple glasshouse crossing experiments can be used to confirm the association of putative S-RNase genotypes with the predicted crossing phenotypes (East & Mangelsdorf, 1925). For example, in SI plants, cross-pollination between individuals with matching S-locus genotypes does not result in fruit set. Informed by such features, we may be able to uncover the genetic basis of SI with transcriptome sequencing and candidate genes, coupled with simple experimental design.

Here, we describe how an easily generalised approach yielded the discovery of RSI in cacti (Cactaceae). Specifically, we employed

common transcriptome-generating methods to find that pistils of *Schlumbergera truncata* express highly polymorphic and obligately heterozygous S-RNases, which are closely related to S-RNases from other plant families with this SI system. Moreover, the inferred S-locus genotypes predict the crossing phenotypes, as assessed by hand pollinations, and we find many pollen-expressed genes that strongly resemble known SLFs. With the recent finding of RSI in Rutaceae (Liang *et al.*, 2020), we add to the weight of evidence that this mechanism is likely to be more common across core eudicots than generally appreciated. A high proportion of cacti – including many rare, threatened and endangered species – express SI (Boyle, 1997; Mandujano *et al.*, 2010). Our candidate-based RNA-seq approach for discovery of this mechanism is simple, inexpensive and easily applied. It holds especially great promise for determining the genetic basis of SI in woody perennials, particularly trees and other species with long juvenile phases. Below, we provide evidence that RSI operates in cacti, explore the resulting insights into the evolution of this widespread SI system, and the clear path to discovery of new instances of SI across flowering plants.

## Materials and Methods

### Study organism

*Schlumbergera truncata* (Haw.) Moran, commonly known as the ‘crab cactus’ or ‘false Christmas cactus,’ is perhaps the most widely cultivated species of cacti (Anderson, 2001). Designated by the International Union for Conservation of Nature (IUCN) as Vulnerable (Taylor & Zappi, 2017), its wild populations have a highly restricted range across the slopes of Brazilian Atlantic forests at the southern edge of the tropics, at elevations of 700–1000 m. This species is easily grown from cuttings and it has been extensively hybridised across Western Europe, since the early 19th century. It produces a rich array of flower colours and forms (Boyle, 2003). Although there are dozens of named varieties, nearly all commercially grown plants are of ultimately uncertain provenance. Our samples were sourced from a haphazardly collected personal collection (BI; accessions 15H01–15H09). Individuals of this species are generally SI, displaying the gametophytic genetic mode of SI, in which a pistil-expressed genotype recognises (and rejects) a matching pollen-expressed S-allele haplotype (Boyle, 1997). One individual, accession 15H06 was specifically included in the study because it was reported to display self-compatibility (SC). SC is regularly expressed by mutant individuals of otherwise SI species (Brown, 2011; Raduski *et al.*, 2012).

Additionally, we were granted access to one individual of *Matucana madisoniorum* (Hutchison) G.D.Rowley from the Huntington Botanical Gardens and Library (Pasadena, CA, USA; accession no. 42118). We were interested in this species, because it is designated by the IUCN as critically endangered, restricted to a single population within an area no greater than 16 km<sup>2</sup> (Ostalaza & Roque, 2013). S-locus genotyping in such rare species could provide crucial information for conservation efforts, because mate limitation due to S-allele scarcity can drive or reinforce population declines (e.g. Busch & Schoen, 2008).

## RNA sequencing

Pistils (without ovaries), pollen, stem (cladode), and root tissues were collected and immediately submerged in 1.5 ml of RNAlater solution (Invitrogen). Submerged samples were held at room temperature for 30 min and then moved to a  $-80^{\circ}\text{C}$  freezer for storage. Here, c. 100 mg of tissue was ground to a fine powder in 1.5 ml tubes submerged in liquid nitrogen. Total RNA was isolated using the Total RNA Mini Kit (IB47341; Plant kit; IBI Scientific) following the manufacturer's instructions. We assessed RNA concentration and purity using a NanoDrop Lite Spectrophotometer (Thermo Scientific). The 23 samples used in this study were sequenced as part of a larger sequencing effort that consisted of four separate sequencing runs and included additional samples from other plant species. Sequencing libraries were prepared using the KAPA Stranded mRNA-Seq (Roche), and these libraries were sequenced on a single lane of the Illumina HiSeq 4000 or Illumina NovaSeq 6000 platform (paired-end 150 bp reads) at the Duke University Center for Genomic and Computational Biology. The number of resulting read pairs (for the 23 samples presented here) ranged from 4148 932 to 9618 084 with a median of 6363 556 and average of 6293 553 (Supporting Information Table S1).

## Transcriptome assemblies

Raw paired-end Illumina reads were first processed using Rcorrector v.1.0.4 Song & Florea, 2015 to correct for random sequencing errors. Then, reads were trimmed with TRIMMOMATIC v.0.39 (Bolger *et al.*, 2014) to remove any read containing bases with Phred scores lower than 20, low quality reads less than 50 bp long, and any adapter or other Illumina-specific sequences that were still present. The remaining reads were filtered with KRAKEN 2 software (Wood *et al.*, 2019) to remove small and large subunit ribosomal RNA (SILVA database) (Quast *et al.*, 2013) and any contaminating reads (minikraken2\_v2 database). Additionally, we used custom-built databases, derived from RefSeq libraries: UniVec\_Core, viral, mitochondrion, plastid, plasmid, archaea, bacteria, protozoa, human and fungi to minimise the number of contaminating and nonnuclear reads. Only paired reads were used for transcriptome assemblies. *Schlumbergera truncata* filtered reads were combined across all samples into a single RNA-seq data set. We conducted a *de novo* transcriptome assembly using TRINITY v2.8.5 (Grabherr *et al.*, 2011) to generate a single reference transcriptome assembly for *Schlumbergera truncata*. The same assembly protocol was followed for the single pistil sample of *Matucana madisoniorum*.

We identified transcripts of candidate genes (S-RNases and SLFs) and expression control/reference genes (EF: Elongation factor 1- $\alpha$  1; TU:  $\alpha$ -tubulin; and AC: actin) in the transcriptome assemblies with the KAKAPO v.0.7.3-dev pipeline (<http://flightless.one>). Transcript quantification was performed using RSEM (Li & Dewey, 2011). Trimmed mean of M-values (TMM, mean of log-expression ratios (Robinson & Oshlack, 2010)) normalisation factors were calculated using SARTOOLS

v.1.7.3 (Varet *et al.*, 2016) with edgeR v.3.30.2 in R v.4.0.2 (Robinson *et al.*, 2010; R Core Team, 2020).

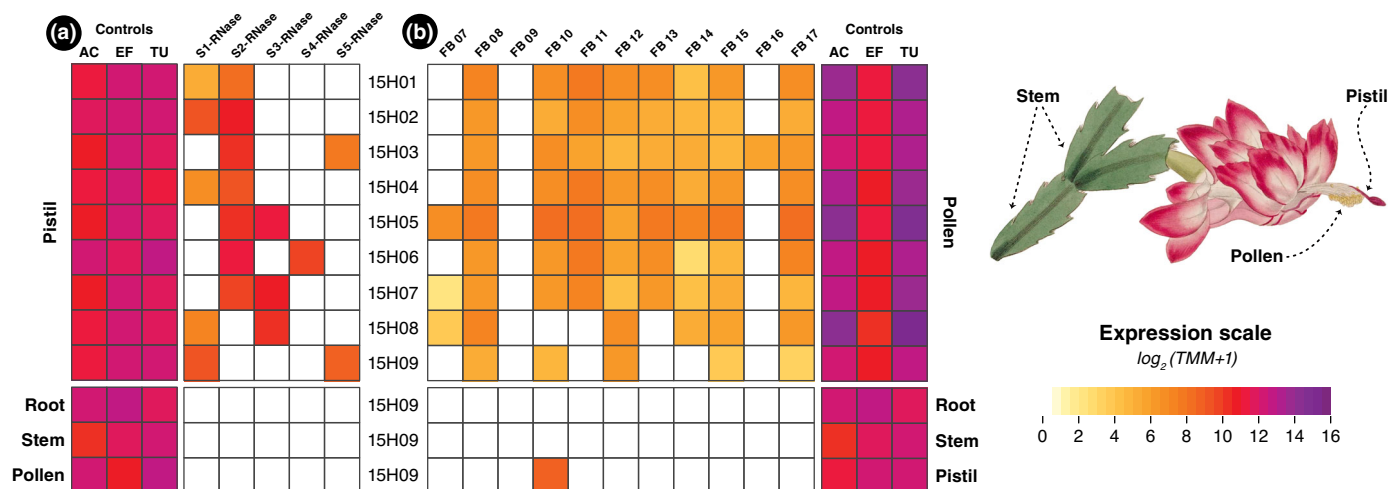
We examined the four published draft genomes of cacti (Copetti *et al.*, 2017), but we only found the S-locus partly assembled in the draft genome of organ pipe cactus, *Stenocereus thurberi*. This is not unusual, because the S-locus is generally difficult to assemble, being heterozygous, highly polymorphic and replete with repetitive elements. We extracted all organ pipe genomic contigs with putative S-RNase or SLF-like sequences and used them to compare intron locations with those inferred from immature mRNA (introns are not spliced in a small fraction of reads) in crab cactus transcriptomes. Finally, we used the PROTEINANALYSIS tool in BIOPYTHON v.1.78 (Cock *et al.*, 2009) to calculate a partly informative and conserved feature of T2/S-RNases, isoelectric point (pI), the pH at which a molecule carries no net electric charge. Signal peptide sequences were not included in pI calculations.

## Phylogenetic analyses

Additional T2/S-RNase and F-box domain-containing sequences were obtained using BLAST by searching the GenBank nucleotide collection (nr/nt) database and the genomes of *Arabidopsis thaliana* (Brassicaceae), *Coffea eugenoides* (Rubiaceae), *Prunus avium* (Rosaceae), *Spinacia oleracea* (Amaranthaceae), *Solanum chilense* (Solanaceae), and *Stenocereus thurberi* (Cactaceae) (Copetti *et al.*, 2017) downloaded from the NCBI FTP server (<ftp://ftp.ncbi.nlm.nih.gov>). The genome of *Citrus maxima* (Rutaceae) (Wang *et al.*, 2017) was downloaded from the 'Citrus Genome Database' website (<https://citrusgenomedb.org>).

Sequence alignments were constructed using DECIPHER v.2.16.1 in R v.4.0.2 (Wright, 2015; R Core Team, 2020). The coding regions of the 453 F-box domain-containing sequences were translation-aligned using the AlignTranslation function. The alignment of the 153 T2/S-RNase sequences was constructed in two stages. First, the coding regions of sequences belonging to Classes I, II and III were aligned separately using the AlignTranslation function, then the three resulting alignments were profile aligned using the AlignProfiles function.

Phylogenetic relationships for T2/S-RNase and F-box domain-containing sequences were inferred using RAXML v.8.2.12 (Stamatakis, 2014) with general time-reversible model and Gamma-distributed among-site rate variation (option -m GTRGAMMA). To evaluate a diverse set of tree topologies and to reduce the chances of getting trapped in local likelihood maxima, the following strategy was used. For each sequence set, RAXML was run 1000 times. For each run, a random starting tree was used (-d), and subtree pruning re-grafting (SPR) moves were randomised (-f t). Additionally, the value for the 'initial rearrangement setting' (-i) was chosen at random (with equal draw probabilities) from a list of values: 2, 4, 6, 8, 12, 16, 20, 24, 32, 48. The tree with the maximum likelihood (least negative log-likelihood score) was selected as the best tree. The log-likelihood scores of the best T2/S-RNase and F-box domain-containing gene trees were  $-56941.30$  and  $-440386.75$ , respectively. Finally, we performed 1000 ordinary (not 'fast')



**Fig. 1** Summary of candidate gene (S-RNase and SLF) expression data for nine *Schlumbergera truncata* individuals. Following tissue-specific transcriptome assembly, we identified putative S-RNase and SLF genes with secondary sequence features and phylogenetic placement (Ramanaukas & Igić, 2017). Individual plant accessions are shown in rows between the two figure panels. Control genes: AC, actin; EF, elongation factor EF-1 alpha; TU, alpha-tubulin. (a) Pistil-expressed putative S-RNase (S<sub>1</sub>–S<sub>5</sub>) follows the expected pattern. Each genotype is heterozygous (expresses two S-RNase alleles). The expression of each predicted allele level is fairly high, in all cases in the top 10% among pistil-expressed genes. Their expression is undetectable in root, stem, and pollen. Subsequent crossing experiments confirmed that these sequences segregate with the S-phenotypes, and successfully predict crossing ability from predicted genotypes. (b) Pollen expression of putative SLFs. We expect at least a dozen SLF genes to be expressed in each S-haplotype, genetically linked with one S-RNase allele, in a functional RSI mechanism (with non-self recognition, Kubo *et al.*, 2010). Each S-haplotype is also expected to express most – if not all – SLF genes, except for the one that recognises its own linked S-RNase. Their expression is undetectable in root, stem, or pistil, except in the case of FB<sub>09</sub> (monophyletic with other putative SLFs, but is not expressed in pollen) and FB<sub>10</sub> (recovered in 15H03 and 15H09 pistils; see text for details). Although this set of genes shows a variety of features broadly consistent with SLFs, we lacked a genome assembly of the S-locus from any Cactaceae species, and our limited expression data include aggregated pollen grains that express the content of both parental S-haplotypes.

nonparametric bootstrap analyses for each dataset (–b). For visual clarity and in view of space considerations, we display only a subsampled T2/S-RNase gene tree in the main body of the paper and relegate the full versions of both gene trees (T2/S-RNase and F-box genes) with hundreds of tip and branch supports (Figs S1, S2).

### Controlled crosses and phenotyping

We performed controlled crosses to confirm the association between an individual plant's SI function (recognition phenotype) and putative S-RNase allele sequences (genotype). Prior data indicated that this species displays gametophytic SI with stylar inhibition (Boyle, 1997, 2003). The molecular genetic basis of the response remained unknown, so our objective centred on determining the genotypes at the putative S-RNases and whether they predicted the crossing phenotypes, given the action of gametophytic SI. For example, we expected crosses with a plant bearing the genotype S<sub>1</sub>S<sub>2</sub> at the S-locus to be reciprocally sterile with all identical genotypes, and reciprocally fertile when crossed with any other genotype (half-compatible crosses yield fruit in functional SI).

Our study plants differed in size and condition (some have ongoing cactus virus × infections), as well as flowering periods, which is common for this species (Boyle, 1997). A full diallel crossing design was not possible, because of the limited number of co-flowering individuals and available flowers. Nevertheless, we self-fertilised each plant and conducted as many cross-pollination as was practical, in all cases using all flowers for either

crossing or tissue collection. For all manual pollinations, we removed *c.* 20 whole stamens (of many more) and rubbed dehiscent anthers over the stigmata, generally on the second or third day after opening. Each resulting cross was labelled with a jewellery tag and scored 14–16 d later (presence of fruit or absence of fruit). As expected, all haphazardly selected individuals displayed SI, fail to set fruit following self-fertilisation. We also found that one plant (15H06), specifically obtained because it was anecdotally reported to be SC, indeed displayed SC. Furthermore, we expected crosses to fail when putative S-allele genotypes matched for crossed individuals (when fully incompatible, as in 15H01/15H02/15H04 & 15H05/15H07), and to succeed when the genotypes were not matched. The authors independently performed genotyping (KR) and phenotyping experiments (BI), and compared the outcomes after both experiments were completed. Finally, we used the assembled transcriptome to confirm that the individuals that shared the S-locus genotypes were distinct genets (unique genetic individuals) and not ramets (clones) by examining their genotypes at single-copy assembled genes (Fig. S3).

## Results

### Pistil expression of S-RNases

Sequencing of *Schlumbergera truncata* pistil RNA resulted in a median transcriptome assembly of 30 842 genes and 51 414 isoforms (range 16 949–42 717 and 24 392–73 047, respectively; Table S1). Most importantly, using noninvasive sampling, relying on as few as one pistil per individual, we reliably recovered



candidate RNases, in total, five distinct sequences that displayed the salient features of functional S-RNases (Ramanauskas & Igić, 2017).

The recovered S-RNase candidates are highly polymorphic and heterozygous (Fig. 1a). The total coding region sequences ranged from 639 to 666 bp, similar to other S-RNases (Table S2). Each contained highly conserved residues, closely resembling the structure of functional S-RNases, including active site histidines present among across the T2/S-RNase protein family. One of the highly conserved features of functional S-RNase allele sequences is their strongly basic isoelectric point. Each candidate we recovered was predicted to be strongly basic (pI range 8.66–9.02), within the range of those expressed in other families with RSI. The gene structure, however, displayed an interesting departure. It contained two introns, of which one was unique among operational S-RNases (Fig. 2, Class IIIA). Intriguingly, this intron pattern is present in several genes (*c.* 4–6) found in soybean and cotton. RSI may operate in those families (Fabaceae and Malvaceae), but has not been characterised to date (Ramanauskas & Igić, 2017). Moreover, the position was shared with non-S-RNase paralogues (Fig. 2, Class ID), although that group possessed an additional intron at the 5'-end (Ramanauskas & Igić, 2017). While the first intron position was shared by nearly all members of this protein family (Fig. 2; position 5), a second one was absent among known S-RNases, but common among non-S-RNase proteins in this gene family (position 9). S-RNases from *Prunus* also possessed a unique intron (position 1), and it seems that intron gains and losses, although relatively rare and reasonably phylogenetically informative, are more common than previously thought (Igić & Kohn, 2001). Finally, the five putative S-RNases were both highly expressed and tissue specific. Each S-allele we recovered was among the top 10% most highly expressed genes in the pistil, but appeared entirely absent in other tissues examined (pollen, petal, stem and root; Fig. 1a).

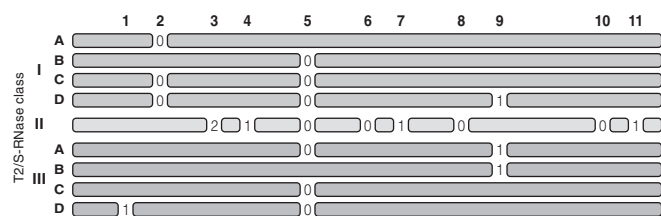
We additionally examined the pistil transcriptome of a single individual of *Matucana madisoniorum*, as well as the published genome (Copetti *et al.*, 2017) of *Stenocereus thurberi* (Sthu\_v1.3,

GenBank assembly accession: GCA\_002740465.1). In both individuals, also members of Cactaceae, we recovered two putative S-RNases apiece, with expression and/or sequence features that closely matched those recovered from *Schlumbergera truncata*. *Stenocereus thurberi* alleles, named S<sub>1</sub> and S<sub>7</sub>, were found on short assembled genome contigs NCQT01146464.1 (2776 bp) and NCQT01107209.1 (39 932 bp), respectively. The allele S<sub>7</sub> was found at the 5'-end of the contig and, based on the alignment with the S<sub>1</sub> allele, it was missing *c.* 93 bp at the 5'-end of the molecule. Each sequence shared the same intron pattern (IIIA). As we did not confirm their functional identity in subsequent phenotyping experiments, we continued to refer to the genes we uncovered in *Matucana* and *Stenocereus* as 'putative' S-alleles or S-RNases, reflecting a considerably lower collective weight of evidence.

S-RNase sequences are expected to display extreme persistent polymorphism, the result of strong and persistent negative frequency-dependent selection (Wright, 1939; Ioerger *et al.*, 1990). We found that pairwise amino acid sequence identity among five *Schlumbergera truncata* S-RNase alleles ranged from 50.5% to 70.1%. This is well within the range found in more intensely sampled species, for example *Solanum chilense* (Solanaceae) in which amino acid identity ranged between 30–96% (Igić *et al.*, 2007). Despite the small sample of alleles, as expected, we found a well supported pattern of trans-generic polymorphism. One of the two *Matucana madisoniorum* S-RNase alleles was closely related (95.3% identical) to *Schlumbergera truncata* allele S<sub>3</sub>, and the other was most similar to the *Schlumbergera truncata* allele S<sub>2</sub> (72.9% AA sequence identity; allele named S<sub>6</sub>). Similarly, one of the two *Stenocereus thurberi* S-RNase alleles was closely related to *Schlumbergera truncata* allele S<sub>1</sub> (96.6% identical), while the other was most similar to the *Schlumbergera truncata* allele S<sub>5</sub> (67.3% AA sequence identity; allele named S<sub>7</sub>).

### Pollen expression of putative SLF genes

Pollen RNA-seq assembly of *Schlumbergera truncata* resulted in a median transcriptome assembly of 28 580 genes and 46 638 isoforms (range 18 571–35 689 and 27 729–58 662, respectively; Table S1). It was comparable to the assemblies recovered in *Petunia inflata* pollen (23 731 and 25 522 genes; Williams *et al.*, 2014). Plant genomes can contain hundreds of F-box domain-containing genes (Gagne *et al.*, 2002; Xu *et al.*, 2009), but only a small fraction of these is expressed in pollen. We restricted our search of F-box domain-containing genes by using only previously described S-locus F-box sequences as BLAST queries. Among the assembled genes, we recovered, in total, 20 transcripts that contained F-box motifs and identified 10 as putative SLFs (see SLF section below). The coding sequence regions of these 10 genes ranged from 1230 to 1341 bp, similar to other SLFs (Table S3) and none were found in other tissues, with the exception of FB\_10, which was recovered from the pistils of 15H03 and 15H09 (Fig. 1b; Table S4). It is unclear, however, whether this was a result of incomplete tissue specificity or contamination. *Schlumbergera truncata* makes copious amounts of pollen and we



**Fig. 2** Intron positions in S-RNases from Cactaceae, relative to the other members of the plant T2/S-RNase gene family. This gene family found across land plants is generally divided into three phylogenetic classes (I–III), largely congruent with intron presence–absence patterns at 11 intron positions (listed along the top). S-RNases from the cacti display pattern IIIA. They contain two introns: one at overall position 5 (phase 0; not interrupting a codon), which is common for this gene family, and another at position 9 (phase 1; interrupting a codon after the first position), previously unknown in functional S-RNases, but shared with Class I T2/S-RNases. Other S-RNases characterised to date display the intron patterns IIIC (many families), and those from *Prunus* (Rosaceae) display the pattern IIID (see Fig. 3 in Ramanauskas & Igić, 2017).

did not emasculate plants used in our experiments, because we collected all pollen and style from the same flowers. The rest were expressed exclusively in pollen with maximum TMM ranges of 59–315 and 0–8 in pollen and pistils, respectively (Table S4).

### S-RNase genotypes predict crossing phenotypes

A functional link (prediction of crossing success), that is the confirmation of an association between putative S-RNase genotypes and the expected crossing phenotypes, is a critical piece of evidence for inference of RSI in *Schlumbergera truncata*. In each case, crossing results with SI genotypes were consistent with the expectation that the RNase sequences we obtained represented functional alleles under single-locus gametophytic SI. The only exception was the sole self-compatible (SC) genotype ( $S_2S_4$ , plant 15H06).

We recovered relatively few alleles (and incompatible groups) in our sample of *Schlumbergera truncata*, five alleles (six S-genotypes) from nine unique individuals (Fig. 1). This relative dearth of diversity at the S-locus is unsurprising. Although crab cacti are commonly grown as ornamental plants, they appear to have their origins in cultivation from relatively few progenitor plants (Boyle, 2003). Such restriction of unique genotypes proved helpful for confirming crossing relationships (Figs 3, S4).

|                | SI             |                |                |                |                |                |                |                | SC             |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| ♀ \ ♂          | S1/S2<br>15H01 | S1/S2<br>15H02 | S1/S2<br>15H04 | S2/S3<br>15H05 | S2/S3<br>15H07 | S2/S5<br>15H03 | S1/S3<br>15H08 | S1/S5<br>17H09 | S2/S4<br>15H06 |
| S1/S2<br>15H01 | ✗              | ✗              | ✗              | ✓              | ✓              | ✓              | ✓              | ✓              | ✓              |
| S1/S2<br>15H02 | ✗              | ✗              |                |                | ✓              |                |                |                |                |
| S1/S2<br>15H04 | ✗              |                | ✗              | ✓              | ✓              | ✓              |                | ✓              | ✓              |
| S2/S3<br>15H05 | ✓              | ✓              | ✓              | ✗              | ✗              | ✓              |                | ✓              | ✓              |
| S2/S3<br>15H07 | ✓              | ✓              | ✓              | ✗              | ✗              | ✓              | ✓              | ✓              |                |
| S2/S5<br>15H03 | ✓              |                | ✓              | ✓              | ✓              | ✗              |                | ✓              | ✓              |
| S1/S3<br>15H08 |                |                |                | ✓              |                |                | ✗              |                |                |
| S1/S5<br>17H09 | ✓              |                | ✓              | ✓              | ✓              |                |                | ✗              | ✓              |
| S2/S4<br>15H06 |                |                | ✓              | ✓              |                |                |                |                | ✓              |

**Fig. 3** Crossing relationships among individuals whose genotypes were identified by RNA-seq. Crossing between dam (rows) × sire (columns) combinations with identical genotypes ought to result in fruit set failure (pollen alleles are recognised and rejected if they match alleles in maternal genotype). When the parental plants possess nonidentical genotypes, crosses ought to be successful. Successes are check-marked, failures cross-marked and empty squares were not attempted. The individual with genotype  $S_2S_4$  (15H06) was included in the study because it is self-fertile (circled). Our data match genotype–phenotype mapping expectations, indicating that the identified sequences are most likely to correspond to functional S-alleles.

All crosses involving SI individuals with matching S-allele genotypes failed (15H01, 15H02 and 15H04 vs 15H05 and 15H07). None of these individuals are clones, each is a unique genet (Fig. S3). Conversely, all crosses involving individuals with differing S-allele genotypes succeeded, albeit with variable rates of fruit set and seed set. Some putatively compatible pollinations failed to produce fruit, as is common in plant studies.

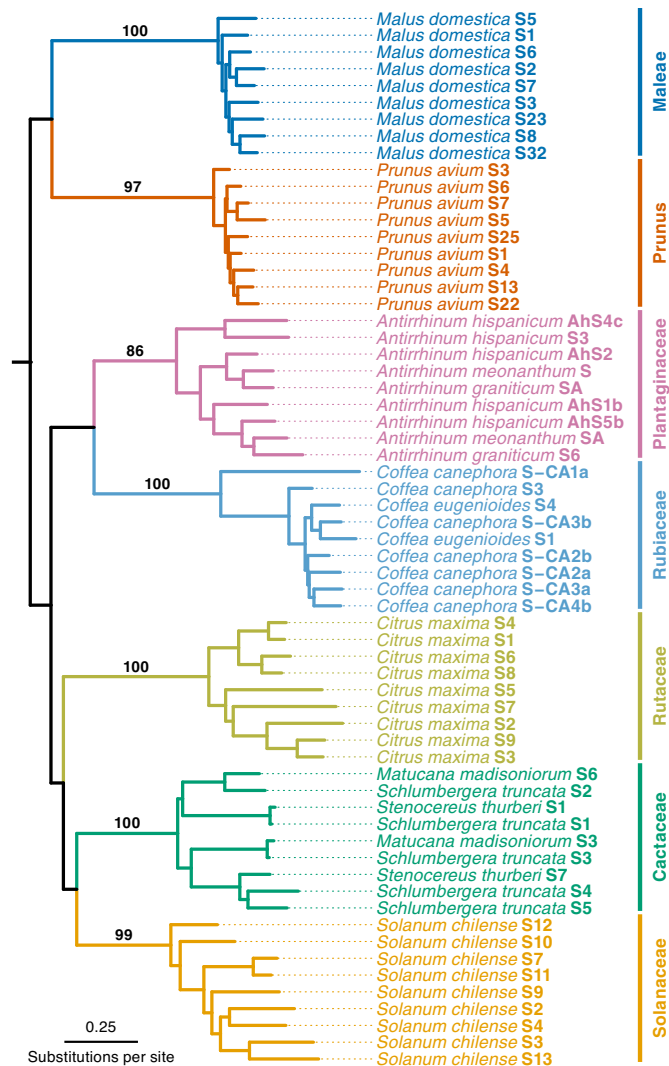
Except for 15H06 – specifically selected because of a reported occurrence of SC – all other individuals were SI (no other individuals set fruit following self-fertilisation). This SC plant is also the only one that expressed S-RNase allele  $S_4$  (genotype  $S_2S_4$ ; Figs 1, S5). While this is a possibility of some consequence, we do not have additional evidence to causally implicate this allele in the loss of SI function. The expression level of this S-RNase allele was within the range of functional alleles and it did not possess readily identifiable loss-of-function mutations (e.g. loss of RNase activity, premature stop codons, frameshift mutations). Taken together, our data point to the possibility that the breakdown of SI in this case may be caused by pollen-part mutations or other genes involved in SI function, which warrants further work. Interpretation of these data is tempered by the fact that we recovered the smallest transcriptome assembly from this SC individual. We found approximately half of the genes/isoforms in 15H06 pollen compared with the median pollen transcriptome assembly. The plant also had the only apparently symptomatic cactus virus × infection.

### Phylogenetic relationships among S-RNases

Gene phylogenies may not form strictly bifurcating relationships when recombination and gene conversion cause homoplasy, especially over 100-Myr time scales (Ramanauskas & Igić, 2017). Nevertheless, information regarding gene relationships among putative S-RNases, especially at shallower time scales, appears to be reasonably well preserved. This idea is supported by comparisons with data independent of DNA sequence alignments used in phylogenetic inference, such as intron presence/absence and isoelectric points (Igić & Kohn, 2001). Therefore, we also examined the inferred phylogenetic relationships among our set of candidates to accumulate additional weight of evidence that these genes were functional S-RNases.

As in nearly all previous studies, our sample of members of the T2/S-RNase family involved in causing pistil-part SI response formed a closely related ‘class’ of RNases, termed ‘Class III RNases,’ which shared conserved structural features and isoelectric points (Table S2). To date, all members of this protein class uncovered within each plant family are monophyletic (Figs 4, S1), although this inference is complicated by the presence of paralogous pseudogenes, loss-of-function mutants and deep divergences.

Even our small sample of S-RNases from *Schlumbergera* and putative S-RNases from *Matucana* and *Stenocereus* is sufficient to recover S-alleles from different genera that are more closely related to one another than to alleles within the same genus. This common S-locus feature, termed trans-generic polymorphism is caused by the action of negative frequency-



**Fig. 4** A maximum-likelihood gene tree of S-RNases. Previously characterised alleles from Solanaceae, Plantaginaceae, Rubiaceae, Rutaceae and Rosaceae are shown alongside Cactaceae S-RNase alleles uncovered in the present study. Alleles from *Schlumbergera*, *Matucana* and *Stenocereus* display trans-generic polymorphism, a common feature of SI systems. (A gene tree displaying a larger sample of plant T2/S-RNase gene family members, along with bootstrap branch support is available in Supporting Information Fig. S1; see text for analysis details.)

dependent selection, a form of diversifying selection on mating type loci (Ioerger *et al.*, 1990). Such selection preserves ancestral polymorphism, which has been uncovered in all cases involving multiallelic SI systems recovered so far. Moreover, trans-generic polymorphism provides unparalleled evidence that the RSI mechanism has been transmitted in an unbroken line of ancestry, from the most recent common ancestor to the extant species (Goldberg *et al.*, 2010) within all families where it is found in eudicots, now including Cactaceae. Finally, the finding of RSI in Cactaceae expands the known distribution of this mechanism to Caryophyllales, an order of plants distantly related to rosids (Rosales and Sapindales) and asterid lineages (Solanales, Lamiales, Gentianales), where it is known to operate.

## Phylogenetic relationships among putative S-locus F-box-containing proteins

Plant genomes contain up to dozens of T2/S-RNases, but hundreds of genes with the N-terminal F-box motif (Xu *et al.*, 2009). This necessarily complicates the detection of putative SLFs and related functional inference (Xu *et al.*, 2013; Williams *et al.*, 2014, 2015). The SLFs across plant families do not form a single clade, with many SLF-like genes breaking the monophyly. Within families, however, SLFs form largely monophyletic groups. We found a similar pattern in the 20 *Schlumbergera truncata* pollen-expressed F-box domain-containing genes, 11 of these (FB\_07–FB\_17) belonged to a clade of closely related sequences. Except for FB\_09, they were nearly exclusively expressed in pollen (Table S4).

Although each S-haplotype was expected to contain many linked F-box containing genes, expression levels of SLFs were much lower than S-RNases, and our extraction methods and coverage were relatively low. Consequently, we cannot confidently infer linked sequences (e.g. haplotype subtraction) and leave it for future work. The 10 putative SLFs also appear highly tissue specific, except for FB\_10, which was also found in pistil RNA. With these limitations in mind, all mutually incompatible individuals with identical S-RNase genotypes (15H01, 15H02 and 15H04 vs 15H05 and 15H07) showed similar putative SLF expression (Fig. 1b). Together with their counterparts from *Stenocereus thurberi*, these genes formed a monophyletic group (Fig. S2). We suspect that extensive gene conversion and recombination among these tandem-duplicated paralogues may undermine the assumptions of a strictly bifurcating underlying process, and yet they also appeared to be relatively closely related to SLF genes in other families.

## Discussion

The combined evidence from RNA-seq assemblies, analyses of gene expression, phylogenetic relationships and crossing data indicated that *Schlumbergera truncata* enforces SI with a genetic mechanism found across eudicots. SI is found in dozens of genera in the cactus family (Boyle, 1997). It is likely to be caused by the same RSI mechanism in most (if not all) instances of SI within the family, as is the case in other plant families expressing RSI. The molecular components of SI are often conserved on smaller phylogenetic scales and members of each family usually rely on the same mechanism (Igić *et al.*, 2008). The site and genetic mode (gametophytic expression) of pollen rejection has been studied in only a few species of cacti, and in each case it is generally consistent with RSI (Boyle, 1997; Martinez-Peralta *et al.*, 2014).

Discovery of putative S-RNase alleles in *Matucana madisoniorum* and *Stenocereus thurberi*, and the preserved ancestral polymorphism implies that this mechanism was also continually present since the common ancestor of nearly all Cactaceae. Transmission of such ancient polymorphism in an unbroken line of descent to present-day species is expected under persistent negative frequency-dependent selection (Igić *et al.*, 2004). The resulting pattern of preserved trans-generic polymorphism among



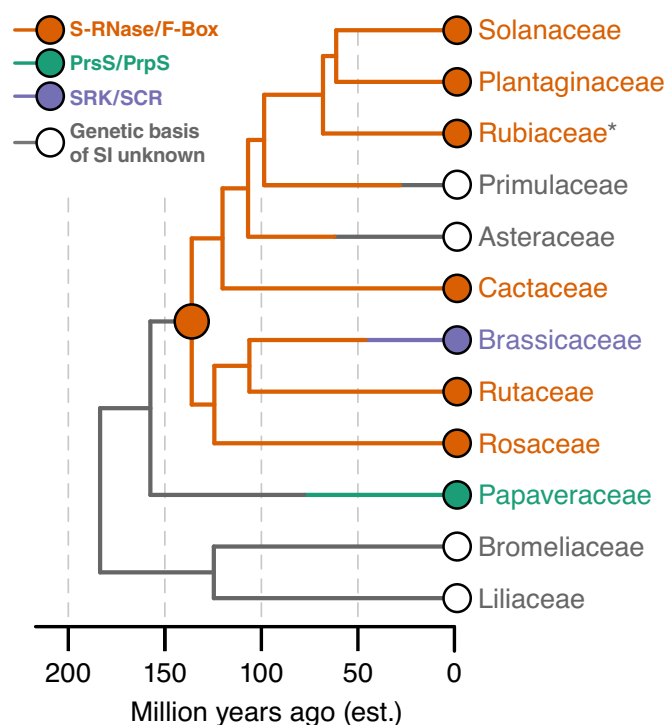
S-RNase alleles reflects the constant presence of SI. This pattern is common within all plant families known to express RSI, and it can be broadly utilised in the study of paleopopulation demography and macroevolutionary inference (Ioerger *et al.*, 1990; Igić *et al.*, 2004).

The finding of RSI in the sixth eudicot family (Fig. 5), is also consistent with the hypothesis that the common ancestor of nearly all eudicots utilised RNases as the pistil component of an ancestral SI mechanism (Igić & Kohn, 2001; Steinbachs & Holsinger, 2002). Cactaceae, which represent one of the most spectacular arid-adapted radiations on Earth, are members of Caryophyllales, an order of over 10 000 species of core eudicots (Soltis *et al.*, 2005). Some members, such as Cactaceae and Chenopodiaceae, have been long known to express gametophytic SI, while others (Caryophyllaceae, Nyctaginaceae) express sporophytic SI (Boyle, 1997). Therefore, it appears that both the conservation of the RSI system as well as the propensity for the origin of novel, nonhomologous, systems had been previously underestimated. In our view, it is all but certain now that a similar and homologous system, with orthologous RNase and F-box-containing specificity-determining components, operated more than 100 Ma.

Cementing the widespread acceptance of the view that RSI is homologous depends on answers to some key questions that could furnish supporting or opposing evidence. First, are there any other homologous SI systems that plausibly span across the same ancestral nodes? For example, discovery of SRK/SCR-like orthologues that control a mechanism in Cactaceae plausibly homologous to the one that operates in Brassicaceae would have weakened or fatally harmed the case for ancestral and homologous RSI (hypothetically). Second, is there another deeply conserved role for the pistil and pollen components that would make them easy to re-recruit in SI or allow function in other roles? If expression of Class III T2/S-RNases were simultaneously involved in an alternative function (Rojas *et al.*, 2018), especially a strongly conserved one, they could serve in such a role. Although a wide variety of functions has been ascribed to genes that belong to this clade of RNases, none has been found to be remotely as conserved as their pistil-expressed role in RSI (Ramanauskas & Igić, 2017). Interestingly, we have previously found that several species of Brassicaceae and Asteraceae (both lacking RSI expression) also appeared to lack Class III T2/S-RNases related to ones in RSI (Ramanauskas & Igić, 2017), which renders deep conservation of other function(s) highly unlikely. Moreover, many other species with non-RSI systems exist and remain unstudied. Third, how widespread and common is RSI? Each new confirmation of the prediction that SI is widespread in eudicots with its discovery in new groups makes hypotheses of re-acquisition increasingly untenable. Finally, given the current state of the art, what is the propensity for change in the molecular genetic components of this system? While it is clear that this mechanism displays considerable variation across families where it is found, we do not understand its capacity for change over vast time scales. For instance, polyploidisation seems to obligately cause the breakdown of SI in most species with RSI, including our focal species *Schlumbergera truncata* (Miller & Venable, 2000; Parks & Boyle, 2003), but a few exceptions exist,

particularly among the plant family Rosaceae (reviewed in Stone, 2002; Mable, 2004). Furthermore, it operates in most species as a non-self-recognition system (pollen-part SLF genes defend the pollen tube by defusing all S-RNases except those within their own haplotype), including the apple genus *Malus* (Kubo *et al.*, 2010). But in closely related stonefruits (genus *Prunus*), orthologous molecular components instead cause self-recognition (pollen-part SLF genes defend the pollen tube by defusing S-RNases linked to their own haplotype), which is a change of genetic and operational significance (Fujii *et al.*, 2016). Whether the mechanism in cacti proceeds by self- or non-self-recognition remains unclear, for now, but the increased pace of discovery of genes that underpin RSI in this and other families clearly marks the beginning of an exciting and accelerated phase in the comparative study of the causes and consequences of breeding system variation in plants.

A baseline expectation for a high propensity for divergence among homologous expression of RSI can be gleaned from previous studies, as well as our present findings concerning *Schlumbergera truncata*. For decades, a suite of superficial traits were commonly used as proxies for molecular mechanisms of SI. Families with trinucleate pollen and dry stigmas, for example, were thought unlikely to express gametophytic SI (Brewbaker, 1967).



**Fig. 5** The summary of the known distribution of SI systems in selected eudicots and monocots. Coloured branches mark a plausible ordering of the evolution of SI systems (key molecular components as identified in the legend). A steadily increasing weight of evidence shows that RNase-based SI is homologous across most core eudicots, and that it was used by the common ancestor of roughly two-thirds of all land plants. This inference yields a prediction – RNase-based SI should be present in a wide swath of eudicots – which we used in our candidate gene approach. The asterisk next to Rubiaceae denotes that some species employ heterostyly, and their molecular basis of SI, as in Primulaceae, is presently unknown.



But these traits fail to accurately predict either the genetic mode or molecular mechanism in Cactaceae. Specifically, cacti exhibit dry papillate stigmas and trinucleate pollen, both of which are instead associated with the sporophytic genetic mode of SI (Brewbaker, 1967; Boyle, 1997). We find RSI in precisely such a family, in apparent departure from this strong angiosperm-wide correlation. At least *Echinopsis*, *Hatiora*, *Rhipsalidopsis*, and *Schlumbergera* exhibit the gametophytic mode (Boyle, 2003), and we find it effected by RSI in *Schlumbergera*. S-RNase alleles from Cactaceae also possess an intron not found in other S-RNases characterised to date. Therefore, even in the absence of a complete understanding of this system in cacti, we can conclude that the RSI system can broadly undergo a great deal of alteration in its constituent components and associated traits.

Our approach paves the search for RSI in other families of core eudicots. Sampled genomes from both Fabaceae and Malvaceae contain genes with the same intron–exon structure as the alleles we recovered from Cactaceae, which are also phylogenetically closely related to other genes known to function in the RSI response (Ramanauskas & Igić, 2017). Although potentially useful, the genomes were from SC species in each case, and such observations consequently provided a reflection of historical, rather than currently operating mechanisms (Ramanauskas & Igić, 2017). Multiple (some non-SI) mechanisms may operate in legumes to lower seed set after self-fertilisation and the majority of species are SC (unpublished data), so the number of candidate RSI species and confidence in its expression in Fabaceae is fairly low (Aguiar *et al.*, 2015b). Ericales and Lamiales are prime candidates for detection of RSI, because many species across their constituent families express SI mechanisms whose properties superficially resemble those found in RSI (Igić *et al.*, 2008). Overall, the total number of species across the core eudicots in which this mechanism is likely to operate has a vast ceiling, of the order of tens of thousands of species. Conversely, however widespread among the eudicots, the RSI mechanism is highly unlikely to be found outside this group. There is no evidence, to date, that any other species (e.g. monocots or magnoliids) express Class III T2/S-RNases (Niu *et al.*, 2017; Ramanauskas & Igić, 2017).

We do not provide the benchmark standard evidence used in molecular biology, an *in vivo* functional assay, for determination of causal relationship between genotype and phenotype. Recombinant S-RNase inhibition of incompatible pollen tube growth *in vivo* or *in vitro* would strongly reinforce our findings, as would the similar generation of pollen tube growth following introduction of recombinant SLF genes (Sijačić *et al.*, 2004), and yet our approach has some important advantages.

Many plants are ill-suited for classical molecular genetic approaches because their juvenile phases may extend over a decade or longer. Even species relatively amenable to manipulation, such as the widely cultivated *Schlumbergera*, require 2 years or more from germination to flowering (Boyle, 1997). Our RNA-seq-based characterisation and correlation of genotype and phenotype expression is simple, inexpensive and easily applied. The present findings are directly relevant for hundreds of species of cacti in precarious existence, where it can be deployed as a part of effective germplasm conservation programme (Boyle, 1997;

Mandujano *et al.*, 2010). More generally, it can be applied in many other families and holds great promise for determining the genetic basis of SI in understudied long-lived perennials, especially tropical trees and many species with long juvenile phases. The general approach can lower the barriers to – and greatly accelerate the pace of – discovery of operational SI mechanisms across angiosperm families, far above the present two families per decade. In principle, coupling a similar informed and targeted experimental design with transcriptome sequencing, even in the absence of candidate genes, may enable the discovery of the genetic basis of entirely novel SI mechanisms. Whatever the case, the indirect manner of such discovery ensures that initial phases of study will require functional assays and detailed elucidation of the underlying mechanism to broaden our understanding of plant breeding system evolution.

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## Author contributions

KR and BI designed the research; KR and BI performed the research; KR and BI conducted data analyses; KR and BI collected the data; KR and BI were involved in data interpretation; KR and BI wrote the manuscript.

## ORCID

Boris Igić  <https://orcid.org/0000-0002-6302-8030>

Karolis Ramanauskas  <https://orcid.org/0000-0002-3513-0695>

## Data availability

This article contains Supporting Information Figs S1–S5 and Tables S1–S4. All sequence data are deposited in GenBank under project accession no. PRJNA705387 (<https://www.ncbi.nlm.nih.gov/Traces/study/?acc=PRJNA705387>). Assembled candidate genes, as well as other key data and scripts will be available on GitHub (<https://github.com/karolis/ramanauskas-igic-2021a>).

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## Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

**Fig. S1** Gene tree of T2/S-RNase gene family.

**Fig. S2** Gene tree of the F-box domain-containing gene family, including pollen-part determinants involved in RSI (SLF).

**Fig. S3** Relationships among individual plants showing that accessions used in analyses are not clones.

**Fig. S4** Crossing relationships among individuals whose genotypes were identified by RNA-seq.

**Fig. S5** Amino acid alignment of Cactaceae S-RNases. Intron positions are marked with triangles.

**Table S1** RNA-seq and assembly summary statistics.

**Table S2** Assembled *Schlumbergera truncata* T2/S-RNase transcripts.

**Table S3** Assembled *Schlumbergera truncata* SLF candidates.

**Table S4** RNA-seq expression data for candidate gene families from *Schlumbergera truncata*.

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